F78AA Actuarial and Financial Mathematics A - Sample solutions to December 2017 Exam

1. (a) The annual interest rate is $1.02^2 - 1 = 0.0404$ effective. As

$$\ddot{s}_{\overline{10}|}^{0.0404} = 12.51435 \quad \text{and} \quad (I\ddot{s})_{\overline{10}|}^{0.0404} = \frac{\ddot{s}_{\overline{10}|}^{0.0404} - 10}{1 - (1.0404)^{-1}} = 64.75069,$$

the accumulated value is $2(I\ddot{s})_{\overline{10}|}^{0.0404} = 2 \times 64.75069 = £129.501$ million.

(b)

$$\begin{aligned}
\ddot{a}_{\overline{n+k}|} =& 1 + v + v^2 + \cdots + v^n + v^{n+1} + \cdots + v^{n+k-1} \\
=& 1 + v + v^2 + \cdots + v^n \left(1 + v^1 + \cdots + v^{k-1}\right) \\
=& \ddot{a}_{\overline{n}|} + v^n \ddot{a}_{\overline{k}|} \\
=& \ddot{a}_{\overline{n}|} + {}_{n|}\ddot{a}_{\overline{k}|}.
\end{aligned}$$

2. Working in pence, the value of the dividend stream at the purchase date is

$$\begin{aligned}
& 1.08^{\frac{7}{12}} \left(20v + 20 \times 1.06 \times v^2 + 20 \times 1.06^2 \times v^3 + \cdots\right) \\
=& 1.08^{\frac{7}{12}} \times 20v \left(1 + 1.06v + (1.06v)^2 + \cdots\right) \\
=& 1.08^{\frac{7}{12}} \times 20v \, \ddot{a}_{\overline{\infty}|}^{0.018868} \\
=& 1.08^{\frac{7}{12}} \times 1\,000 = 1\,045.92.
\end{aligned}$$

3. Let P be the price that the investor paid per £100 nominal. Then, as CGT will be payable ,

$$\begin{aligned}
P =& 4 \times 0.8 \times a_{\overline{10}|}^{0.06} \times 1.06^{\frac{8}{12}} + [100 - (100 - P) \times 0.2] \times 1.06^{-9\frac{4}{12}} \\
\Rightarrow \left[1 - 0.2 \times 1.06^{-9\frac{4}{12}}\right] P =& 3.2 \times 7.36009 \times 1.06^{\frac{8}{12}} + 80 \times 1.06^{-9\frac{4}{12}} \\
P =& \frac{24.48520 + 46.44104}{0.88390} \\
=& 80.24261.
\end{aligned}$$

4. (a) i.

$$(1 + \text{TWRR})^2 = \frac{106}{100}\frac{122}{116}\frac{135}{126} \quad \Rightarrow \quad \text{TWRR} = 0.09291$$

ii. Let i_1 be the MWRR in 2015, with $100(1+i_1) = 106$, i.e. $i_1 = 0.06$.
Let i_2 be the MWRR in 2016, with

$$116(1+i_2) + 4(1+i_2)^{0.5} - 135 = 0$$

Solving the quadratic,

$$(1+i_2)^{0.5} = \frac{-4 \pm \sqrt{16 + 4(135)(116)}}{2(116)} = 1.0616888 \quad \Rightarrow \quad i_2 = 0.1271831$$

Therefore,

$$(1 + \text{LIRR})^2 = 1.06 \times 1.1271831 \quad \Rightarrow \quad \text{LIRR} = 0.09308$$

(b) LIRR = TWRR when the sub-intervals chosen for the LIRR coincide with the intervals between net cash flows being received.
In this instance, the sub-intervals would need to be:

- 1 Jan 2015 – 31 Dec 2016,
- 1 Jan 2016 – 30 June 2016, and
- 1 July 2016 – 31 Dec 2016.

5. (a) The monthly interest rate is $j := 1.02^{\frac{1}{3}} - 1 = 0.006623$ and the annual interest rate is $i := 1.02^4 - 1 = 0.082432$. Let P represent the initial repayment. Then P satisfies

$$\begin{aligned} 200\,000 =& P a^{j}_{\overline{120|}} + (P+200) \times (1+j)^{-120} \times a^{j}_{\overline{180|}} \\ \Rightarrow P =& \frac{200\,000 - 200 \times (1+j)^{-120} \times a^{j}_{\overline{180|}}}{a^{j}_{\overline{120|}} + (1+j)^{-120} \times a^{j}_{\overline{180|}}} \\ \Rightarrow P =& \frac{200\,000 - 200 \times (1+j)^{-120} \times a^{j}_{\overline{180|}}}{a^{j}_{\overline{300|}}} \end{aligned}$$

As

$$a^{j}_{\overline{300|}} = 130.1493,$$

then

$$P = \frac{190\,491.5820}{130.1493} = 1\,463.64.$$

(b) Capital repaid on 1 February 2010 is

$$1\,463.59 - j \times 200\,000 = 1\,463.59 - 0.006623 \times 200\,000 = 138.99.$$

(c) Capital outstanding after the monthly repayment due on 1 January 2017 has been made is

$$\begin{aligned} & P a^{j}_{\overline{36|}} + (P+200) \times (1+j)^{-36} \times a^{j}_{\overline{180|}} \\ =& 1\,463.59 \times 31.93660 + 1\,663.59 \times (1+j)^{-36} \times 104.97482 \\ =& 184\,440.64 \end{aligned}$$

(d) Let P' represent the new annual repayment. Then P' satisfies

$$P'a^{i}_{\overline{10}|} = 184\,440.64 \qquad \Rightarrow \qquad P' = \frac{184\,440.64}{6.63709} = 27,789.39.$$

6. (a) Discounted payback period only indicates when a project would come into profitability (in PV terms). It does not indicate how profitable the project is, which the NPV does.

(b) $NPV_A = 3 \times 1.05^{-10} - 1 = 0.84173976$, i.e. £841 739.76.
For Project B, $NPV_B = 0.08a^{0.05}_{\overline{10}|} + 0.01\,(Ia)^{0.05}_{\overline{10}|} - 1$. As

$$a^{0.05}_{\overline{10}|} = 7.72173, \quad \ddot{a}^{0.05}_{\overline{10}|} = 8.10782 \quad \text{and} \quad (Ia)^{0.05}_{\overline{10}|} = \frac{\ddot{a}^{0.05}_{\overline{10}|} - 10v^{10}}{0.05} = 39.37378,$$

$NPV_B = 0.08 \times 7.72173 + 0.01 \times 39.37378 - 1 = 0.01147662$, i.e. £11 476.62.

(c) The DPP of Project A is 10 years as there are no cash flows before time 10 but project is profitable.

(d) Clearly, the DPP of Project B must be less than 10 years as project is profitable at time 10 and income is received throughout the project time-frame.

(e) Project A is better as it has a higher NPV, even though the DPP is later because the cash flows are weighted towards the end of the project time-frame.